Atomization energy of solids

	DFT	MACE-MP-0	Δ	Δ/DFT [%]
Ag	3.0980	3.0560	0.0420	1.4
Pd	4.3980	4.3560	0.0420	1.0
Rh	6.4040	6.4380	-0.0340	0.5
Li	1.7820	1.7820	0.0000	0.0
Na	1.2440	1.2300	0.0140	1.1
K	0.9870	0.9700	0.0170	1.7
Rb	0.8810	0.8500	0.0310	3.5
Cs	0.8080	0.7770	0.0310	3.8
Ca	2.1380	2.1370	0.0010	0.0
Sr	1.8100	1.8120	-0.0020	0.1
Ba	2.0790	2.0790	0.0000	0.0
Al	3.8910	3.8320	0.0590	1.5
Cu	4.0750	4.0840	-0.0090	0.2
Si	4.9220	4.8680	0.0540	1.1
Ge	4.0360	4.0870	-0.0510	1.3
C	8.0330	7.9910	0.0420	0.5
LiF	4.2670	4.1830	0.0840	2.0
NaF	3.7980	3.7320	0.0660	1.7
NaCl	3.1290	3.1080	0.0210	0.7
MgO	4.7100	4.7020	0.0080	0.2
SiC	5.7620	5.7130	0.0490	0.9
GaAs	2.5480	2.4720	0.0760	3.0
LiCl	3.3770	3.3540	0.0230	0.7
			MAE	
			0.03	

Table S5: Comparison between DFT and MACE-MP-0b3-calculated atomization energies of solids. The column Δ reports the difference between DFT and MACE-MP-0 values. Energies are reported in eV/atom.

Lattice constants of solids

	DFT	MACE-MP-0b3	Δ	Δ/DFT [%]
Ag	4.0820	4.0820	0.0000	0
Al	4.0020	3.9880	0.0140	0
Ba	4.9760	4.9490	0.0270	1
C	3.5620	3.5510	0.0110	0
Ca	5.4630	5.4410	0.0220	0
Cs	6.1060	5.9220	0.1840	3
Cu	3.5680	3.5480	0.0200	1
GaAs	5.6900	5.6760	0.0140	0
Ge	5.7190	5.6770	0.0420	1
K	5.1910	5.1910	0.0000	0
Li	3.3520	3.2890	0.0630	2
LiCl	5.0560	5.0490	0.0070	0
LiF	3.9950	4.0210	-0.0260	1
MgO	4.2030	4.2070	-0.0040	0
Na	4.1070	4.1850	-0.0780	2
NaCl	5.5850	5.5750	0.0100	0
NaF	4.6190	4.5980	0.0210	0
Pd	3.8910	3.9120	-0.0210	1
Rb	5.5720	5.5720	0.0000	0
Rh	3.7600	3.8090	-0.0490	1
Si	5.4340	5.4170	0.0170	0
Sr	5.9080	5.9660	-0.0580	1
SiC(a)	3.0720	3.0780	-0.0060	0
SiC(c)	5.0290	5.0310	-0.0020	0
			MAE	
			0.03	

Table S6: Comparison between DFT and MACE-MP-0b3 lattice constants of solids. The column Δ reports the difference between DFT and MACE-MP-0b3 values. Lattice constants are in Å.

B.5 Reaction barrier heights

In the following section, we benchmark MACE-MP-0 against the reaction barrier heights of the CRBH20 database (*90*), comprising 20 barrier heights for the cycloreversion of heterocyclic rings. The set-up of the DFT calculations is the same as in appendix B.3. MACE-MP-0 and DFT barrier heights (in eV) are reported in table S7. The MAE is approximately 0.3 eV.

Similarity statement

The CRBH20 dataset comprises barrier heights of organic molecules, containing H, O, C, N, S, and F atoms. We report the list of atoms with the number of structures in which they are contained in parentheses: O (73799), S (11972), H (10312), N (11356), F (11277), C (9043). Overall, the database contains no structures matching an exact chemical formula in CRBH20. We provide `crbh20_chemiscope_input.json` to help visualize the interactive UMAP on `chemiscope.org`.

Reaction barrier heights

	DFT	MACE-MP-0b3	Δ	Δ/DFT [%]
1	1.7194	2.0431	-0.3237	19
2	1.9241	1.9988	-0.0747	4
3	1.7499	1.8505	-0.1006	6
4	1.8238	1.8179	0.0059	0
5	1.7237	1.8621	-0.1384	8
6	1.5653	1.1341	0.4312	28
7	1.0911	1.1134	-0.0223	2
8	1.8983	1.6116	0.2867	15
9	1.5477	1.7161	-0.1684	11
10	1.7115	1.1390	0.5725	33
11	1.7379	1.8005	-0.0626	4
12	2.0361	1.6698	0.3663	18
13	1.8739	1.5611	0.3128	17
14	1.9760	1.6166	0.3594	18
15	1.8865	1.5719	0.3146	17
16	1.5741	0.7963	0.7778	49
17	1.2587	0.8127	0.4460	35
18	1.7497	1.3373	0.4124	24
19	1.6989	1.4281	0.2708	16
20	1.7654	0.8742	0.8912	50
			MAE	
			0.3	

Table S7: Comparison between DFT and MACE-MP-0b3 barrier heights for CRBH20. The column Δ reports the difference between DFT and MACE-MP-0b3 values. Energies are in eV.

B.6 Homonuclear diatomics

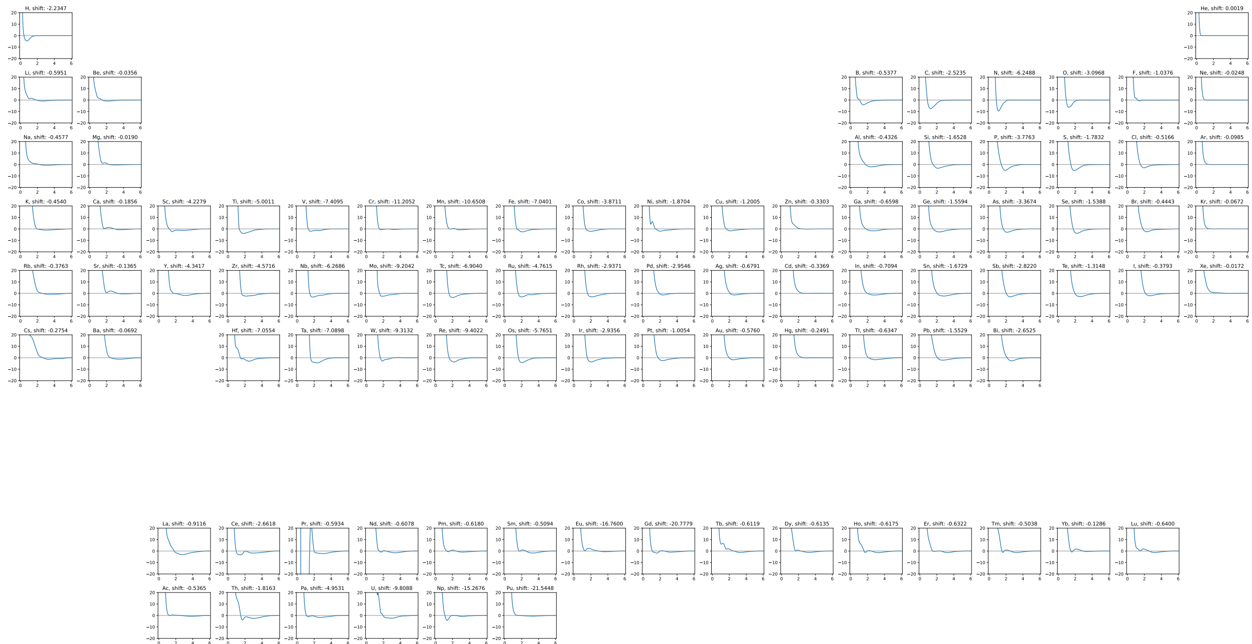

Figure S62: Energies of homonuclear diatomics in vacuum. Interactions are repulsive at small distances for the entire periodic table.

Core repulsion is essential for stable modeling of atomic interactions, so atoms are prevented from coming too close together, especially when modeling high temperatures and pressures. The energies of all pairs of atoms were evaluated with the model in vacuum to test the 2-body interaction. The resultant curves for homonuclear diatomics are plotted in Fig. S62. All elements have repulsive potential at small distances, even elements with minimal presence in the training set (Fig. S66). As an outlier, the Praseodymium pair shows an attractive non-physically large potential with an energy gain of over 450 eV for two Pr atoms combining in vacuum.

C Training Methods and Data Exploration

C.1 Training protocol

The pretrained MACE-MP-0 interatomic potentials consist of many-body message passing (interaction) layers. In each layer, the message is encoded in the irreducible representation basis with C channels up to an angular frequency of order L. This is specified as `(128x0e+128x1o)` for 128 channels and $L = 1$.

In each batch updating step, the weighted sum of Huber losses (*320*) of energy, forces, and stress incurred by all structures in a batch are averaged and back-propagated into the neural networks:

$$
\begin{aligned}
\mathcal{L} = \frac{\lambda_E}{N_b} \sum_{b=1}^{N_b} \mathcal{L}_{\text{Huber}}\left(\frac{\hat{E}_b}{N_a}, \frac{E_b}{N_a}, \delta_E\right) + \frac{\lambda_F}{3\sum_{b=1}^{N_b} N_a} \sum_{b=1}^{N_b}\sum_{a=1}^{N_a}\sum_{i=1}^{3} \mathcal{L}^{\star}_{\text{Huber}}\left(-\frac{\partial \hat{E}_b}{\partial r_{b,a,i}}, F_{b,a,i}, \delta_F\right) \\
+ \frac{\lambda_\sigma}{9N_b} \sum_{b=1}^{N_b}\sum_{i=1}^{3}\sum_{j=1}^{3} \mathcal{L}_{\text{Huber}}\left(\frac{1}{V_b}\frac{\partial \hat{E}_b}{\partial \varepsilon_{b,ij}}, \sigma_{b,ij}, \delta_\sigma\right),
\end{aligned}
\tag{S9}
$$

where $\lambda_E, \lambda_F, \lambda_\sigma$ are predetermined weights of energy, forces, and stress losses. $(\lambda_E, \lambda_F, \lambda_\sigma) = (1, 10, 10)$ is adopted. Huber deltas $\delta_E = \delta_F = \delta_\sigma = 0.01$ are used. In particular, we use conditional Huber loss $\mathcal{L}^{\star}_{\text{Huber}}$ for forces, where the Huber delta δ_F is adaptive to the force magnitude on each atom. To be specific, the Huber delta δ_F decreases step-wise by a factor from 1.0 to 0.1 as the atomic force increases from 0 to 300 eV/Å/atom. The Huber loss for forces can therefore be equivalently represented as:

$$
\mathcal{L}^{\star}_{\text{Huber}}\left(\frac{\partial \hat{E}_b}{-\partial r_{b,a,i}}, F_{b,a,i}, \delta_F\right) = \begin{cases} \mathcal{L}_{\text{Huber}}(\ldots, \delta_F), & F_{b,a,i} < 100 \\ \mathcal{L}_{\text{Huber}}(\ldots, 0.7\delta_F), & 100 \leq F_{b,a,i} < 200 \\ \mathcal{L}_{\text{Huber}}(\ldots, 0.4\delta_F), & 200 \leq F_{b,a,i} < 300 \\ \mathcal{L}_{\text{Huber}}(\ldots, 0.1\delta_F), & F_{b,a,i} \geq 300 \end{cases}
\tag{S10}
$$

Standardization of target variables (here energies, forces and stresses) with different scales has been proven to be important for weight initialization and training stability (*321*), in that a large spread of input or output will result in large and uneven weight values and cause model instability. After each message passing layer k, the node energies ϵ_a are scaled and shifted before sum pooling. The energy prediction of each structure therefore reads:

$$
\hat{E} = \sum_{a=1}^{N}\left[\sigma\left(\sum_{k=1}^{K}\epsilon_a^{(k)}\right) + \mu_{Z_a}\right],
\tag{S11}
$$

where K denotes the total number of message passing layers and $\epsilon_a^{(k)}$ is the node energy of atom a at k-th layer. σ is the root mean square of the atomic forces computed over the training dataset. In order to ensure the correct limit for the dissociation of atoms, we take μ_Z as the isolated atom energy computed with the MPtrj DFT. The predicted forces and stress are computed through PyTorch's automatic differentiation `torch.autograd` of total energy with respect to atomic positions and lattice strain tensor.

The original MACE-MP-0a models were trained for 200 epochs with 40–80 NVIDIA A100 GPUs across 10–20 nodes on HPE (Hewlett Packard Enterprise) Cray EX supercomputer Perlmutter, maintained by National Energy Research Scientific Computing Center (NERSC), a U.S. Department of Energy Office of Science User Facility located at Lawrence Berkeley National Laboratory (LBNL).

The updated MACE-MP-0b3 models is trained for 99 epochs with 32 NVIDIA H100 GPUs across 8 nodes on the Jean Zay cluster managed by the GENCI–IDRIS.

All the finetuned model were trained on a single NVIDIA H100 GPUs Jean Zay cluster managed by the GENCI–IDRIS.